Lecture plan · for the lecturer

Real-time and the power budget

IoT · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Budget energy with duty cycles.
- Define hard and soft real-time.
- Check schedulability.

Activity: Measure, then schedule

1. Warm-up (5 min): أين ذهبت الطاقة؟
2. Measure (20 min): Teams log active and sleep current on their node.
3. Budget (15 min): Compute I_avg and predicted life, then halve the wake time and recompute.
4. Schedule (15 min): Compute U for three tasks and test the rate-monotonic bound.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- How far was the datasheet from your measurement?
- Is your system hard or soft real-time?
- What would you send less often?

Exit ticket

- Compute the battery life of a node at 0.25 mA average on 1,000 mAh.

Try it in your next lecture
Give each team a current meter and its own node; they measure active and sleep current, predict battery life and defend the number.

Seminar sheet · for students

Real-time and the power budget

IoT · University · Lecture 2

Name _____ Class _____ Date _____

1 **Tasks (C = 1 ms, T = 4 ms) and (C = 2 ms, T = 10 ms). Under the 2-task RM bound of about 83%?**

◯ Yes, U = 45% ◯ No, U = 120% ◯ Yes, U = 30%

2 **Work out the battery life.**

Active 30 mA for 2 s every 60 s, sleep ≈ 0. I_avg = _____ mA, so 2,000 mAh lasts _____ hours.

Word bank: 1 · 30 · 60 · 2000

3 **Which changes extend battery life most for a radio sensor? Choose all that apply.**

☐ Send a daily summary instead of every reading

☐ Use the deepest sleep mode between readings ☐ Shorten time-on-air by sending fewer bytes

☐ Run the processor at full speed all the time

4 **A real-time system is one that runs as fast as possible.**

◯ True ◯ False

5 **Match each task to its real-time class.**

Airbag trigger	• • • •	Soft
Video frame display	• • • •	Hard
Pacemaker pulse	• • • •	Soft
Weather dashboard refresh	• • • •	Hard

✎ **How do your students currently estimate power? What would change if they had to measure it?**